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HW Set 02: Root Finding

Kaitlyn Parrinello[1]

[1] *Homer L. Dodge Department of Physics and Astronomy, University of Oklahoma, Norman, OK 73019, USA*

ABSTRACT

The second homework: it's root finding time. All code can be found in the Jupyter Notebook: https://github.com/kparrine/Computational_Astro/blob/main/C2/C2.ipynb.

1. QUESTION 1

The derivations are available through these links:

1. Part a
2. Part b
3. It took 15 iterations for the bisection method (BM) and it took four iterations for the Newton-Raphson (NR) method.

2. QUESTION 2

The results for part (a) in the code are linked in the abstract.
The results for part (b) is summarized in the Table 1. BM is very stable because always making sure that the chosen x values bracket the root. However, it takes many iterations to solve for the root even if I narrow the initial conditions to right around the root, it still takes more than 20 iterations. I noticed that most of my initial guesses took around 30 iterations even if the initial guesses were somewhat close or extremely far.

The results for part (c) is summarized in the Table 2. NR can break or be led astray if the function has local minima/maxima or if the function has a derivative equal to zero. The only time my code actually gave me an error was when I found where the slope is equal to 0 which is in the last line of the table. The most iterations I got was 119, so there must have been some going back and forth before NR got out of a well and got to the root. The relative error calculated here is much smaller than what BM could do on account of the faster convergence.

3. QUESTION 3

My equation of interest is Kepler's equation:

$$M = E - e \sin E$$

where M is the mean anomaly, E is the eccentric anomaly, and e is the eccentricity.

I think this equation can be quite fun for coding and for the goal of root finding in mind. This equation is a transcendental equation (thanks sine), so it can only be solved numerically. In the past, I had done the NR method for Halley's Comet and plotting the x and y positions of its orbit, but that ended up being a much more complex problem since we started from first principles to derive the orbit. With this, I feel like I can scale it back a little and try the more general case.

To go about solving for the NR method, I go ahead and put everything to one side:

$$f(E) = E - e \sin E - M = 0$$

Now I can take the derivative wrt to E:

$$f'(E) = 1 - e \cos E$$

Before I can use the NR method, I need to solve for the mean anomaly, M:

$$M = n(t - t_0)$$

where n is the mean motion, t_0 is the initial time, and t is the time some time after t_0. The mean motion is given by

$$n = \frac{2\pi}{P}$$

where P is the period. With these pieces, I can use the NR method to solve for E. Now, I can also get the x and y positions for any eccentricity with these relations:

$$x = a[\cos(E) - e]$$

$$y = b \sin(E)$$

where a is the semi-major axis and b is the semi-minor axis. Again, the code is linked in the abstract, but the results are in Figure 1. My inputs are $a = 1.0$, $P = 1.0$, and time values are defined by `t_vals = np.linspace(0, P, 300)`. In words, one complete orbit is sampled by 300 points.

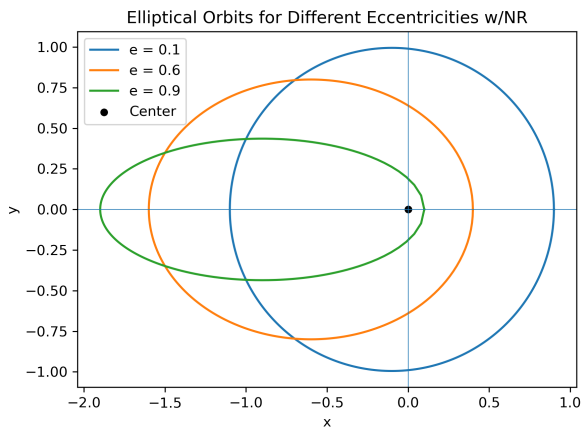

Figure 1. Orbits with different eccentricities of 0.1 (blue), 0.6 (orange), and 0.9 (green). The center is at the origin where a star could be.

I try three eccentricities, 0.9 (almost a circle), 0.6 which is somewhere in-between an ellipse and a circle, and then 0.1 which is elliptical. As expected, these are the shapes that are produced from the function. I notice that as the eccentricity increases, the NR convergence time slows down. For example, for the most part the NR reaches convergence within two to three iterations for $e = 0.1$, but for $e = 0.9$, the convergence happens after about seven iterations. Effectively, as the eccentricity increases, NR takes more time for convergence of the eccentric anomaly.

Table 1. Bisection Method Results

x_0	x_1	# of iterations	Final Relative Error
0	1	27	7.45×10^{-9}
-5	3	30	7.45×10^{-9}
-12	20	32	7.45×10^{-9}
-0.5	0.5	27	7.45×10^{-9}
0.44	0.46	21	9.54×10^{-9}
-150	200	34	5.09×10^{-9}

Table 1. Bisection method results for various initial guesses. The table lists the starting and ending points of each initial guess, x_0 and x_1, respectively, the number of iterations it takes to converge, and the final relative error calculated.

Table 2. Newton-Raphson Method Results

x_0	# of iterations	Final Relative Error
0	5	5.27×10^{-15}
0.43	4	3.39×10^{-10}
1.7	119	2.998×10^{-15}
230	23	6.08×10^{-14}
-2	7	1.72×10^{-10}
10000	80	4.496×10^{-15}
100	90	1.33×10^{-12}
-34.2	13	4.44×10^{-16}
$\frac{7+\sqrt{7}}{3}$	No convergence	nan

Table 2. Newton-Raphson method results for various initial guesses. The table lists each initial guess, x_0, the number of iterations it takes to converge, and the final relative error calculated.